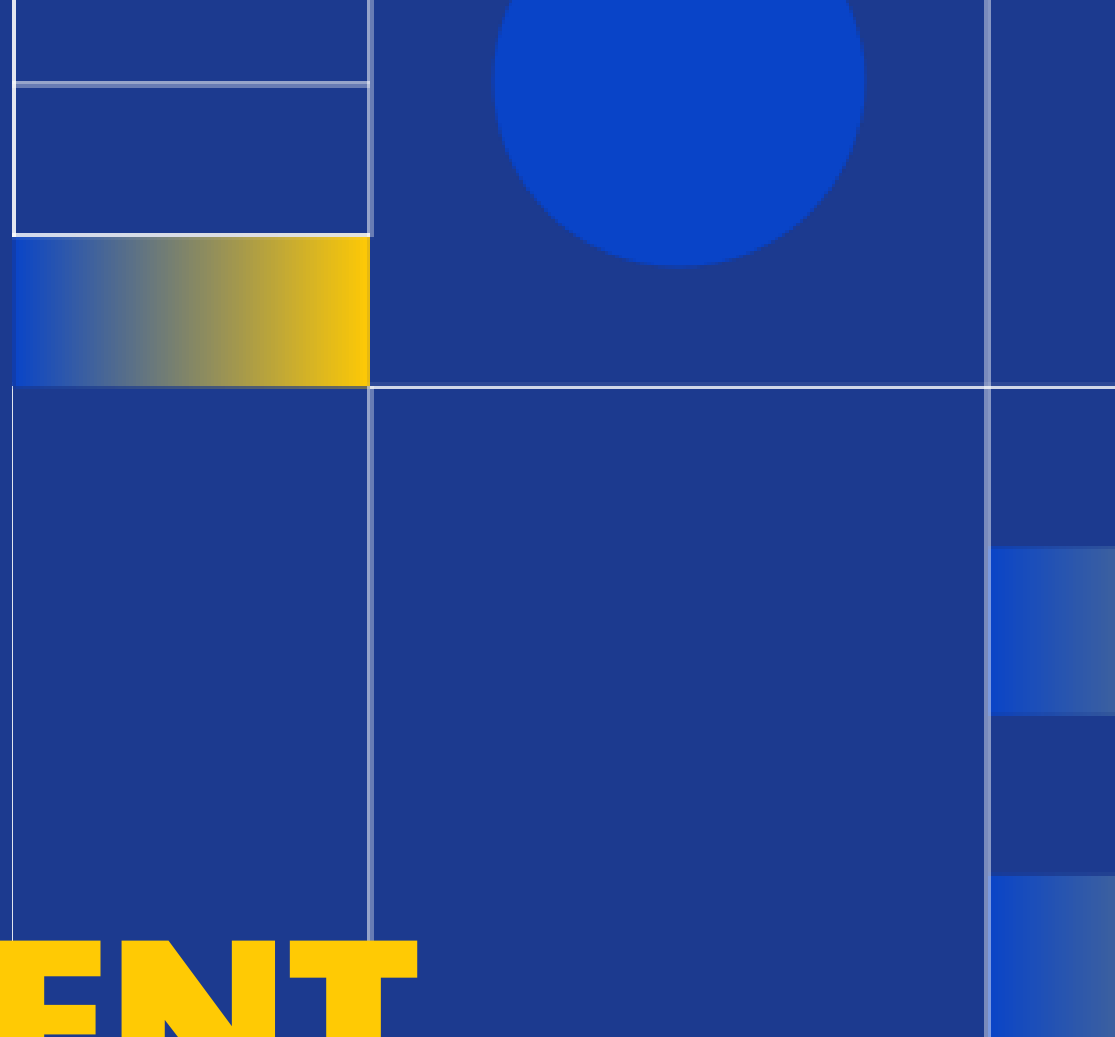
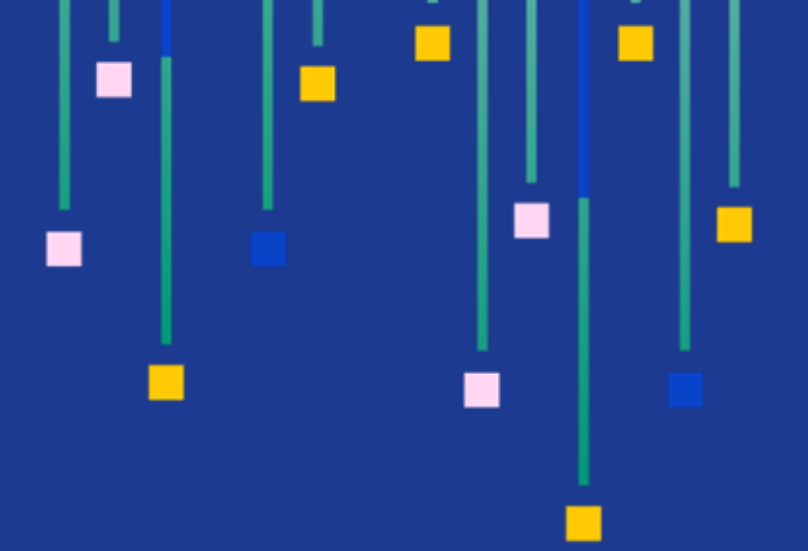
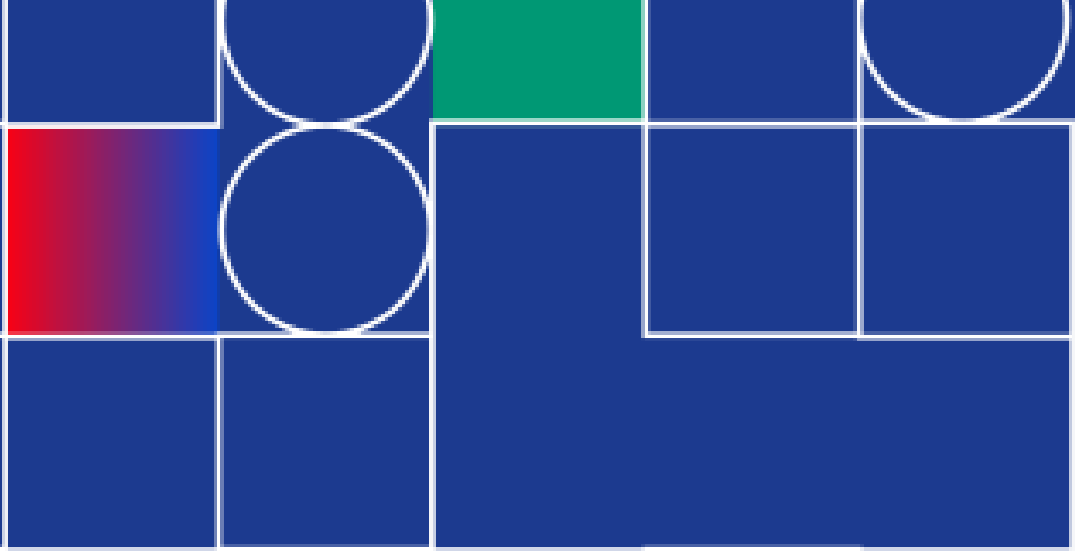




INTRODUCTION TO WEB DEVELOPMENT

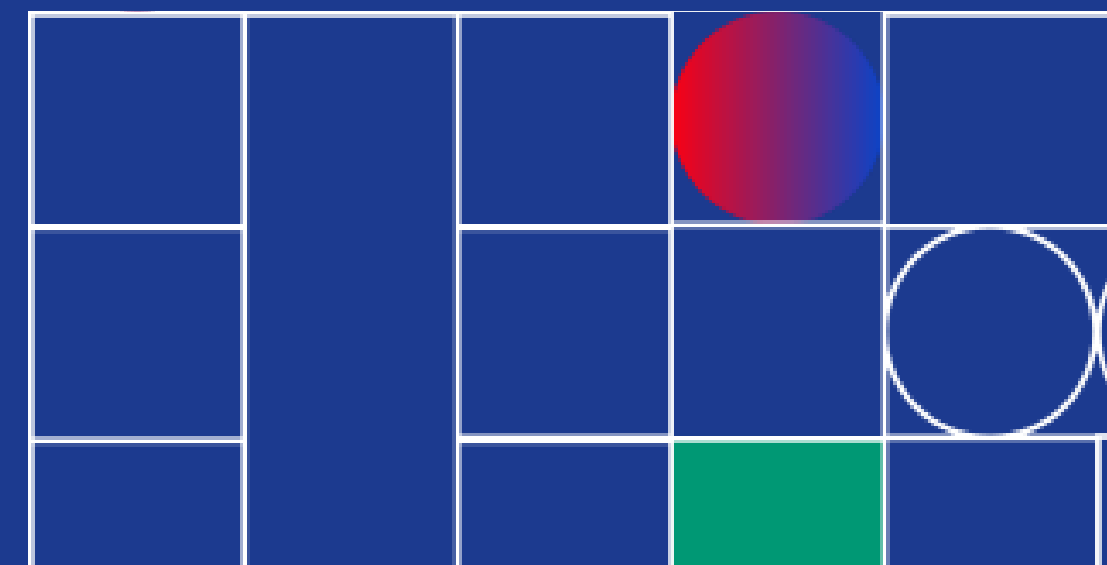
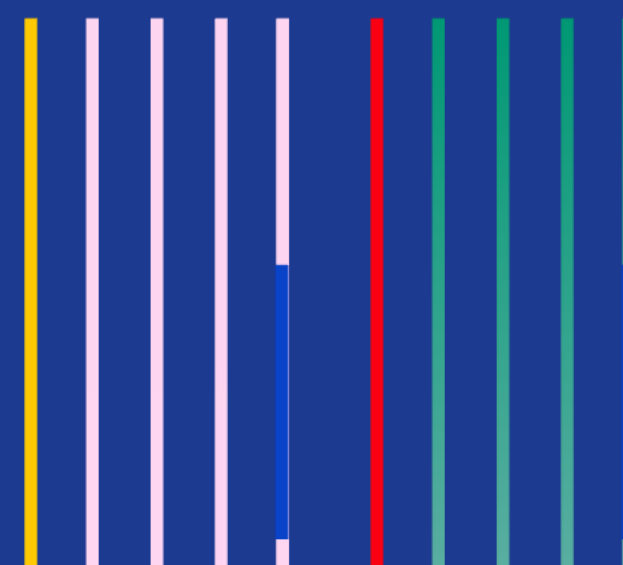
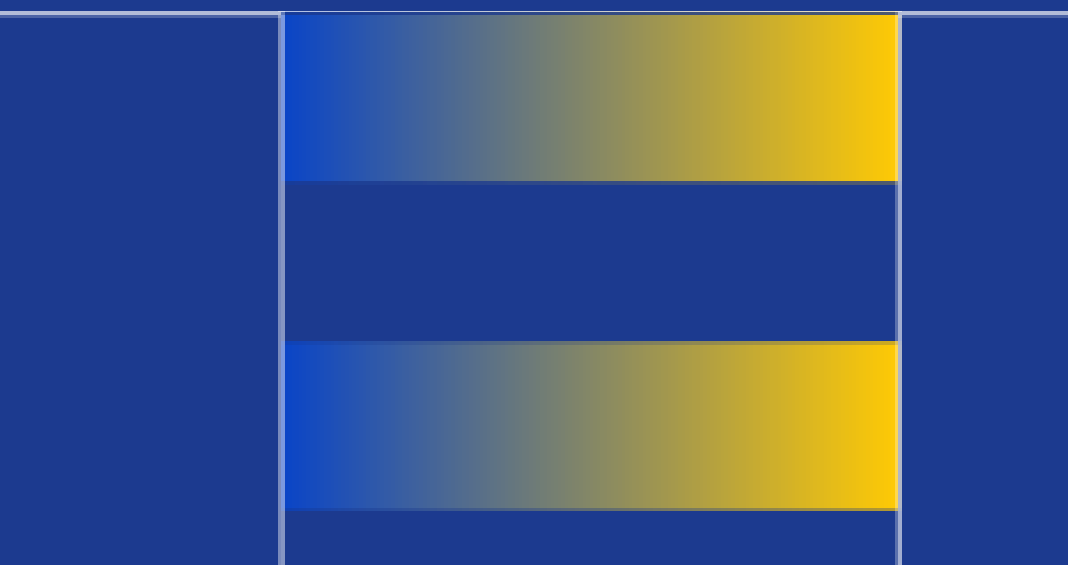


Faculty Mentor
Dr Manonmani S

Presented by :

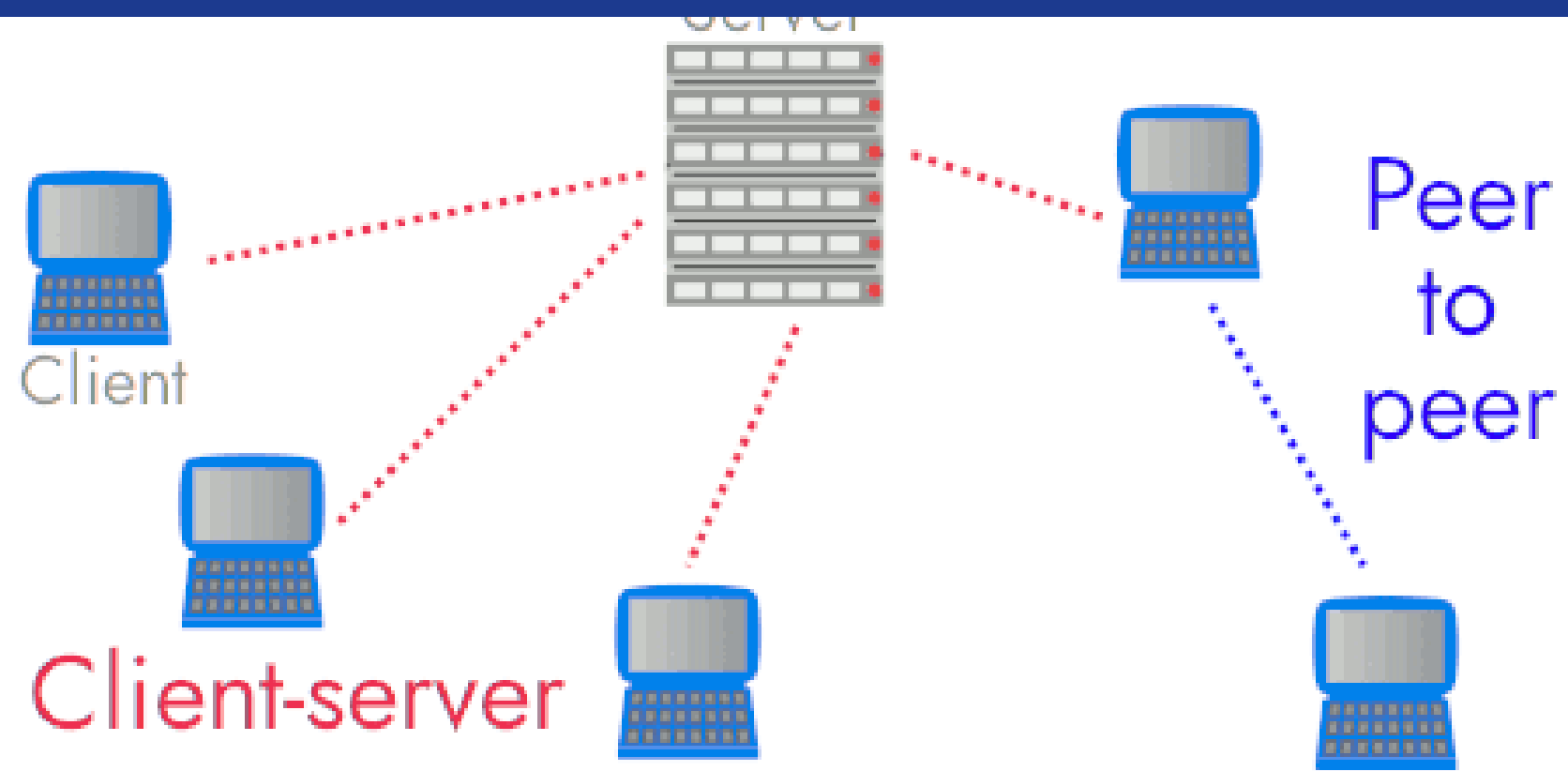
Inika

Kavin Krishnan C



What is internet?

Interconnected network of computers
around the world



```
self.dict_key_name(self) = self  
iter = self.stub.GetResponse()  
  
ses = {}  
for data in resp_iter:  
    status = Status(  
        status_id=data.id, name=  
    )  
    statuses[status.name] = status
```

What happens when you type www.google.com?



What is web development?



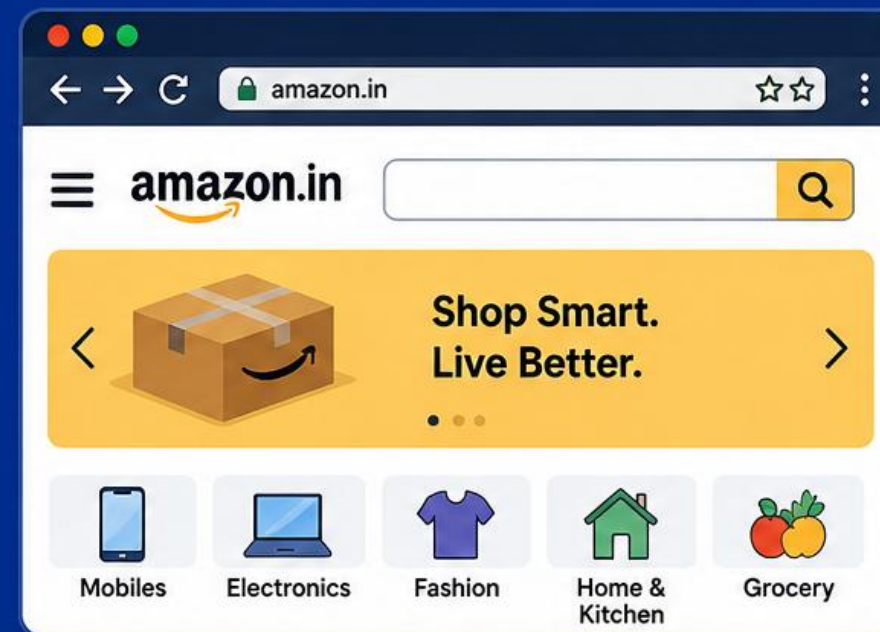
Browser / Client

request

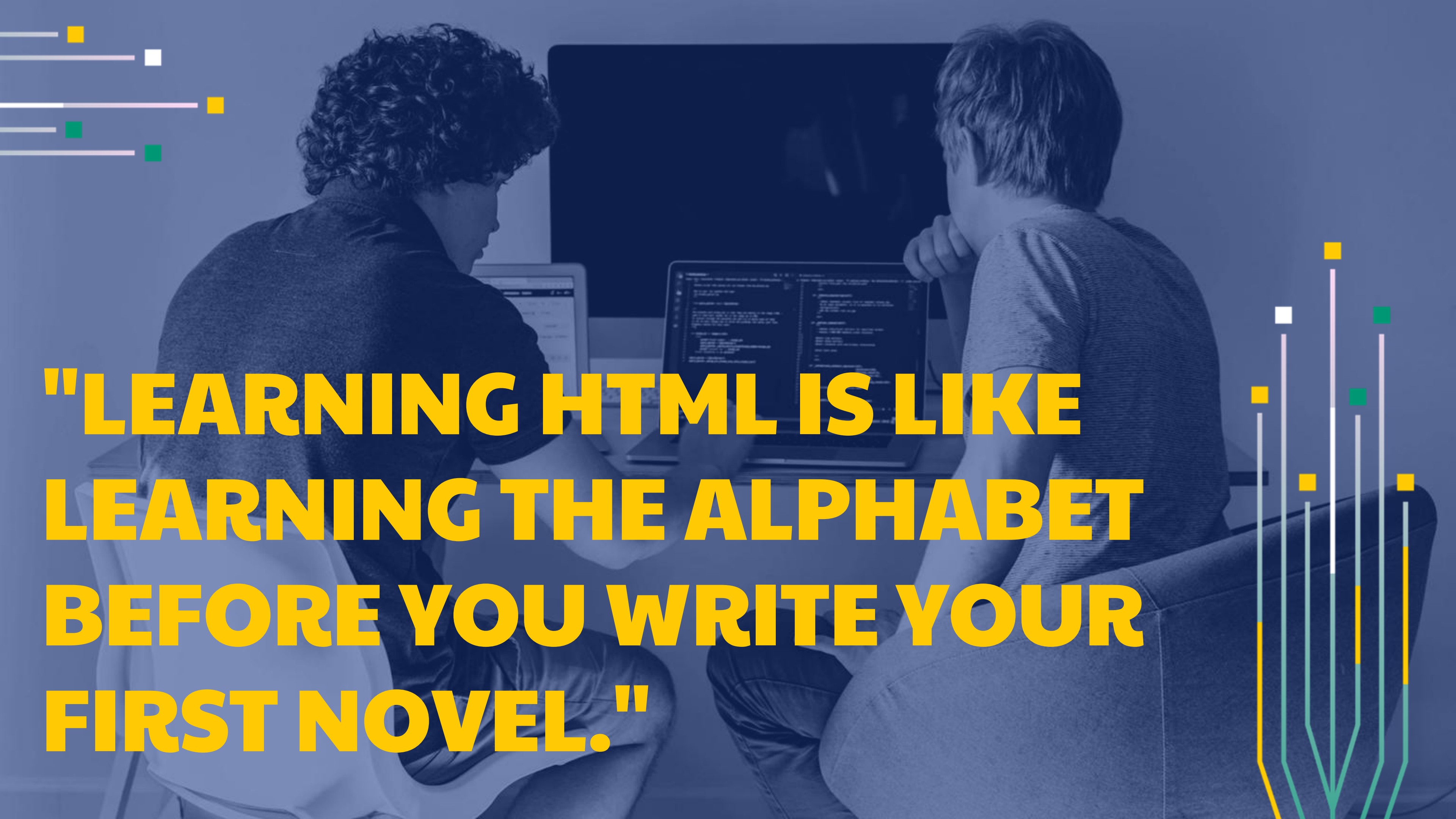


Amazon Server

response



Amazon.in (Website)



**"LEARNING HTML IS LIKE
LEARNING THE ALPHABET
BEFORE YOU WRITE YOUR
FIRST NOVEL."**

HTML STRUCTURE

EXAMPLE

```
<!DOCTYPE html>  
<html>  
  <head>  
    <title>My First Web Page</title>  
  </head>  
  <body>  
    <h1>Hello, World!</h1>  
    <p>This is a simple HTML document.</p>  
  </body>  
</html>
```

Elements & tags

- Paragraph Tag
- Heading Tags – Wikipedia page
- Unordered lists
- Ordered lists

Anchor Tag

- Used to add links to your page
- `<a href = "https://google.com"> Google</a>`

Image Tag

- Used to add images to your page
- `<img src = "dog.jpeg" alt = "dog image" height="100px" width = "100px">`

Quick Question

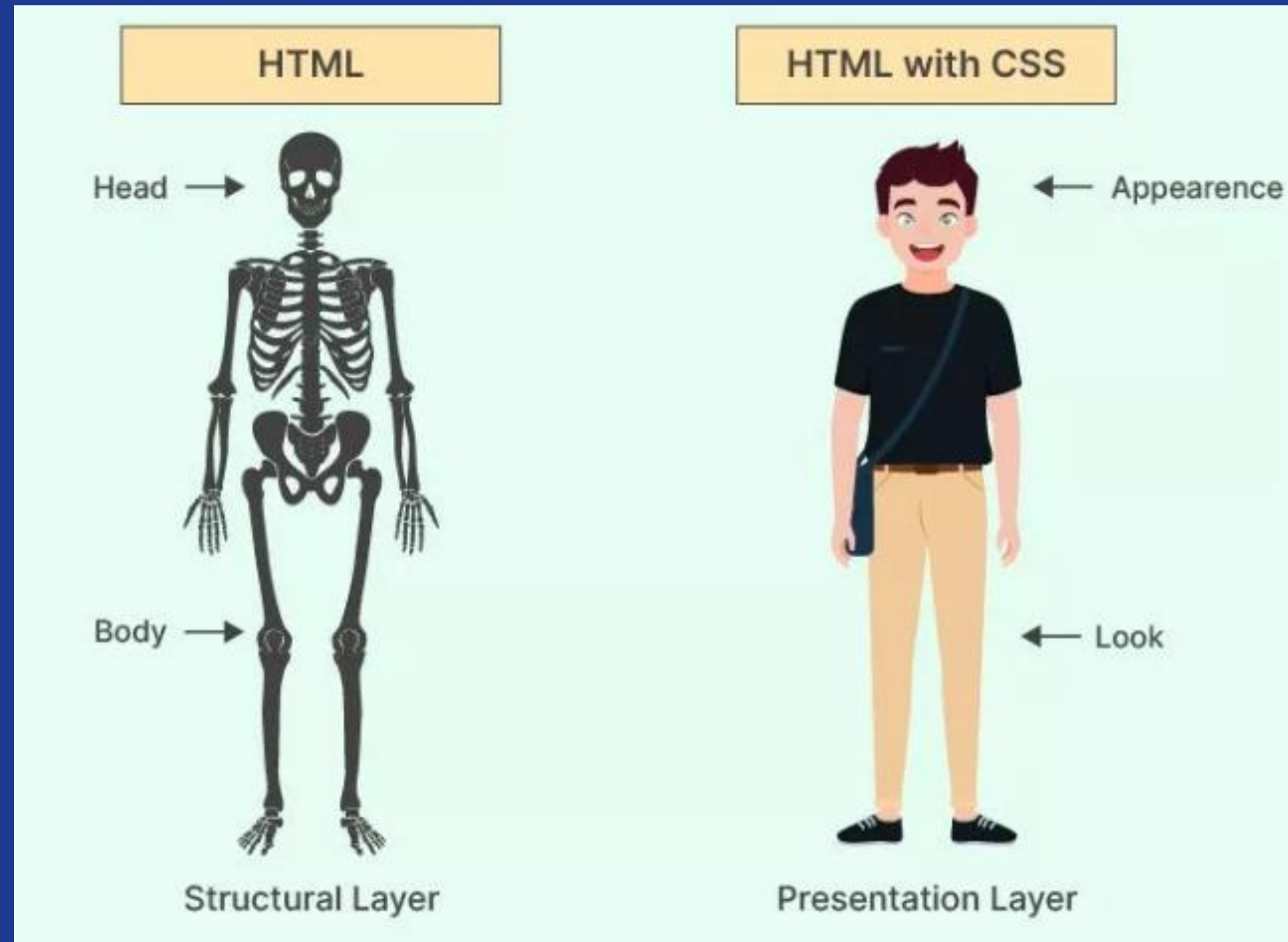
List of Fruits

- Apple [see an apple](#)
- Orange [see an orange](#)
- Mango [see a mango](#)

More Tags

- Anchor tag – inline
- Div container - ref google website
- Span

CSS



To begin with..

- Inspect amazon website...

Inline Style

- `<h1 style = "color : red">`

Welcome `</h1>`

Style tag

- <head>

<style>

h1{

color:red;

background-color : blue;

}

</style>

</head>

Linking HTML & CSS

- <head>
 <link rel = “stylesheet” href =
 “style.css”/>
</head>

Color Property

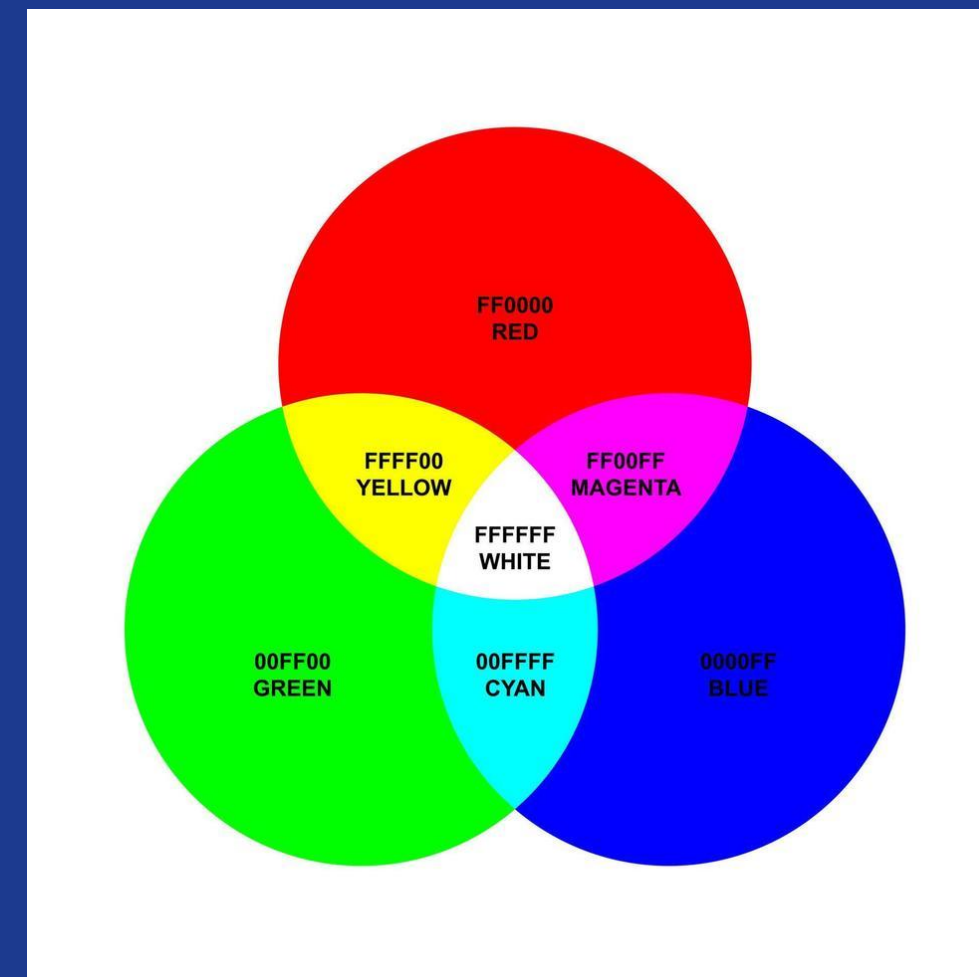
- H1{
 color:red;
}
- P{
 color:blue;
}

BG-Color Property

- H3{
background-color:white;
}
- div{
background-color:black;
}

COLOR CODES

- Hex-codes
- Rgb



Text Properties

- Text-align
- Font-weight
- Text-decoration
- Line-height
- Letter-spacing
- Font-size
- Font family – backup fonts

Selectors

- Universal Selector
- Element selector
- Id selector
- Class selector

!important

- H1{
background-color:blue; !important
}
- h1,h3{
background-color:red;
}

Mini Project



Inika

Second Year, Data Science

Hi! I'm a first-year engineering student who loves coding, music, and badminton. This is my very first web page!

My Interests:

HTML
CSS
C++

Find me here: [GitHub](#) | [LinkedIn](#)

Index.html

```

```

```
<h1 id="main-name">Inika</h1>
```

```
<h2>Second Year, Data Science</h2>
```

```
<p class="bio">
```

Hi! I'm a first-year engineering student who loves coding, music, and badminton. This is my very first web page!

```
</p>
```

Index.html

```
<p><strong>My Interests:</strong></p>
```

```
<ul>
```

```
<li class="tag">HTML</li>
```

```
<li class="tag">CSS</li>
```

```
<li class="tag">C++</li>
```

```
</ul>
```

```
<p>
```

Find me here:

```
<a href="https://github.com" target="_blank">GitHub</a> |
```

```
<a href="https://linkedin.com" target="_blank">LinkedIn</a>
```

```
</p>
```

Style.css

```
*{  
  
  font-family: "Poppins", Arial, sans-serif; /* backup fonts if Poppins isn't installed */  
  
}  
  
/* ---- Element selector: applies to every <body> (just one, but still an element selector) ---- */  
  
body{  
  
  background-color: #f4f6f8;  
  
  text-align: center;  
  
}  
  
/* ---- Element selector: every <h1> ---- */  
  
h1{  
  
  color: #2c3e50;  
  
  font-size: 32px;  
  
  font-weight: bold;  
  
  letter-spacing: 1px;  
  
}
```

Style.css

```
h2{
```

```
  color: #7f8c8d;
```

```
  font-size: 16px;
```

```
  font-weight: normal;
```

```
  text-decoration: underline;
```

```
}
```

```
/* ---- Class selector: any element with class="bio" ---- */
```

```
.bio{
```

```
  color: #333333;
```

```
  font-size: 15px;
```

```
  line-height: 1.8;
```

```
  letter-spacing: 0.3px;
```

```
}
```


Style.css

```
.tag{  
  
  color: #2c3e50;  
  
  background-color: #ecf0f1;  
  
  font-weight: bold;  
  
  font-size: 14px;  
  
  text-decoration: none;  
  
}  
  
/* ---- Element selector: every <a> link ---- */  
  
a{  
  
  color: #2980b9;  
  
  text-decoration: none;  
  
  font-weight: bold;  
  
}
```

Style.css

```
/* ---- Id selector: the one element with id="main-name" ---- */
```

```
#main-name{
```

```
    color: #e67e22;
```

```
    text-decoration: underline;
```

```
}
```

JavaScript

- Lenskart web-page
- REPL (Read-Eval-Print-Loop)
- Using the console

Variables

- A variable is simply the name of a storage location

- Example : a = 10

age = 23

name = John

Data Types

- Number
- Boolean
- String
- Undefined
- Null
- Using typeof
- Example : Instagram profile page

Arithmetic

- Like any other language general arithmetic operations can be performed on JavaScript.

Identifier Rules

- Can contain letters, digits, underscores and dollar signs
- Names must begin with a letter
- Names can also begin with \$ and _.
- Names are case sensitive (a and A are different)
- Reserved words(keywords) cannot be used.

Boolean

- Let age = 13;
- Let isAdult = false;

Strings

- Let sentence = “this is apple”;
- Let sentence = “this is ‘apple’”;
- String indexing – 0 based
 - sentence[0]
 - sentence.length
 - “this” + “ ” + “is” (concatenation)

Null & Undefined

- Null is the intentional absence of any object value
- A variable that has not been assigned a value is of type undefined

console.log()

- To write a message on the console
- Find the difference
- “Hello!”;
- Console.log(“Hello!”);

console.log()

- Let num = 123;
- Console.log("num");
- Console.log(num);
- Console.log(1+2);
- Console.log("Hello", "World", num, (1+5));

Linking JS file

- `<body>`
 `<h1> Hello </h1>`
 `<script src="app.js"></script>`
 `</body>`

Template Literals

- Let pencil = 5;
- Let eraser = 2;
- Console.log("The total cost is:", pencil + eraser, "rupees");
- Let output = `The total cost is \${pencil+eraser} rupees`;
- Console.log(output);

Operators

- Arithmetic (+, -, *, /, %, **)
- Unary (++ , --)
- Assignment (=, +=, -=, *=, /=)
- Comparison
- Logical
- Comparison for non-numbers based on ascii

Conditional Statements

- If-else
- Nested – if-else
- Switch case

If Statements

- If(condition){
 console.log("something");
}
 console.log("After if");

Else - If Statements

- If(condition){
 console.log("something");
}

 else if(condition){
 console.log("anything");
}

 else if(condition){
 console.log("statement");
}

Else Statements

- If(condition){
 console.log("something");
}

 else(condition){
 console.log("anything");
}

Nested - if Statements

- Let marks = 45;
- If(marks >= 33){
 console.log("Pass");
 if(marks >= 80) {
 console.log("Grade : O");
 } else {
 console.log("Grade: A");
 }
} else {
 console.log("Better Luck next time!");
}

Truthy & Falsy

- If(true){
 console.log("It has true value");
}
else{
 console.log("It has false value");
}
- Falsy values – false, 0,-0, "", null , undefined, NaN

Switch - Case

- Let color = "red";
- Switch(color) {
 - case "red": console.log("stop");
break;
 - case "yellow" : console.log("slow down");
break;
 - case "green" : console.log("GO");
break;
 - default : console.log("Broken Light");}

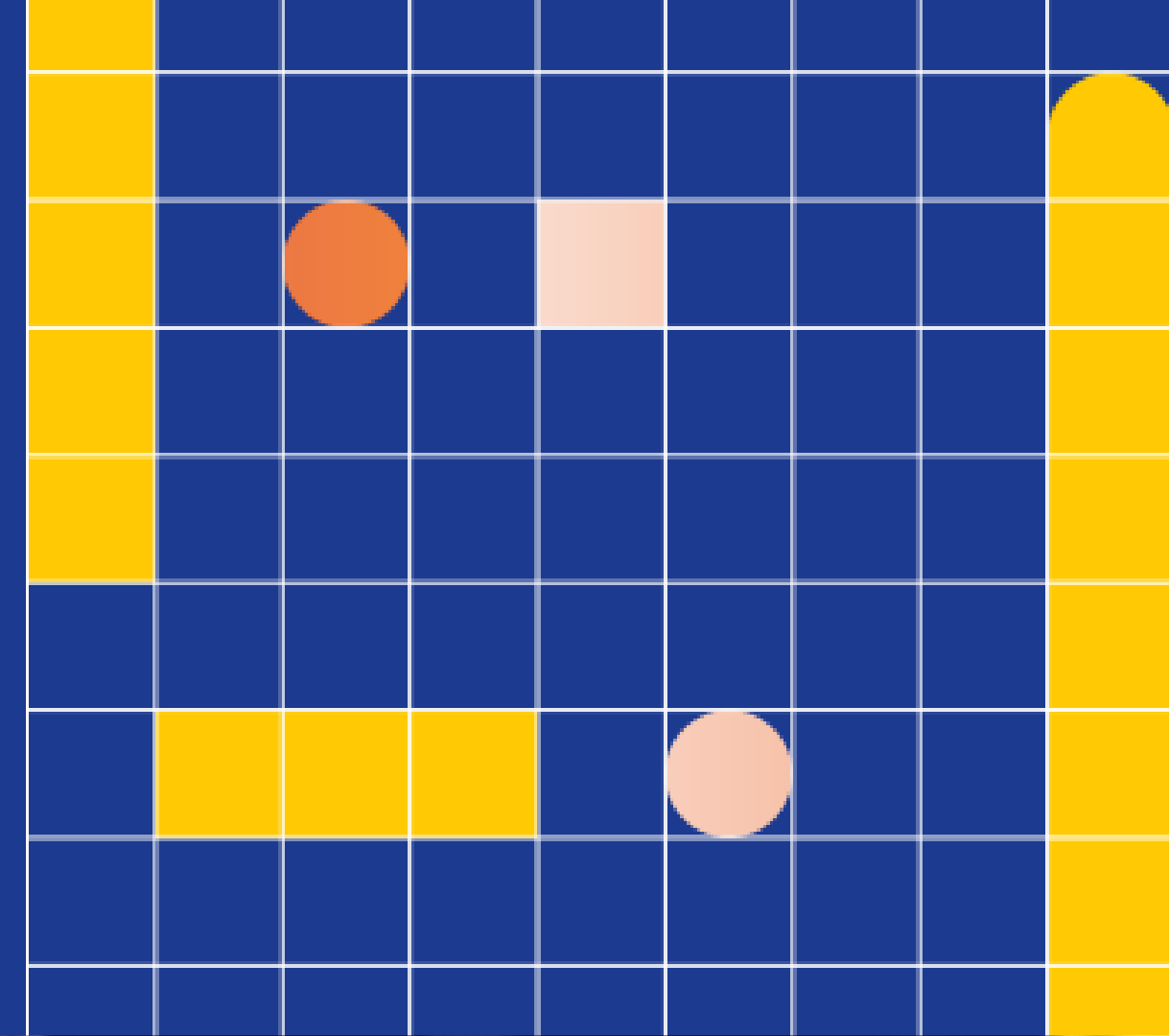
Alert & Prompt

- Alert displays an alert message on the console
- Alert("something is wrong!");
- Prompt displays a dialogue box that asks user for input
- Prompt("please enter your roll no.");

What is git?

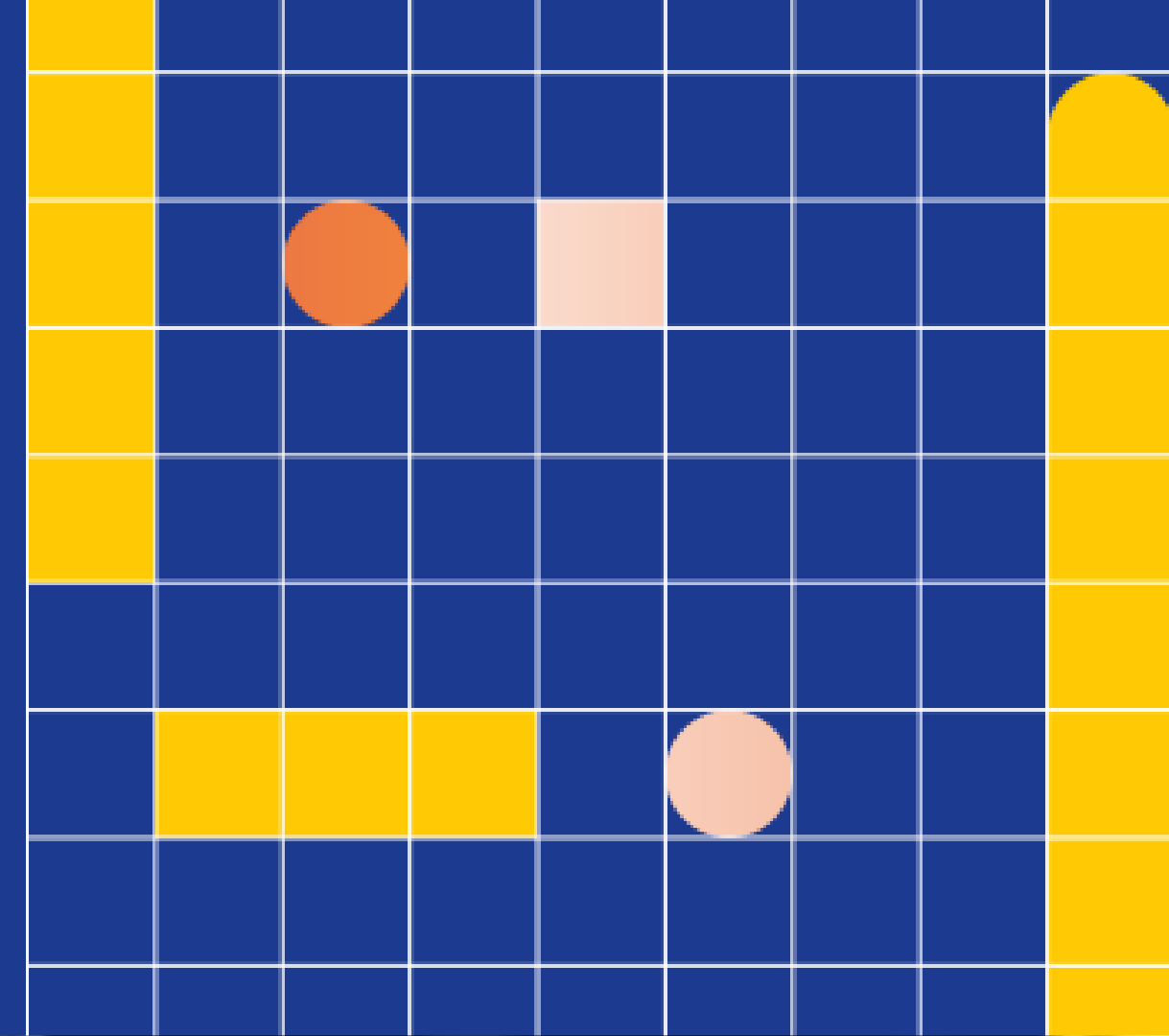
Free and Open source version control system

- Helps to track history
- Helps to collaborate



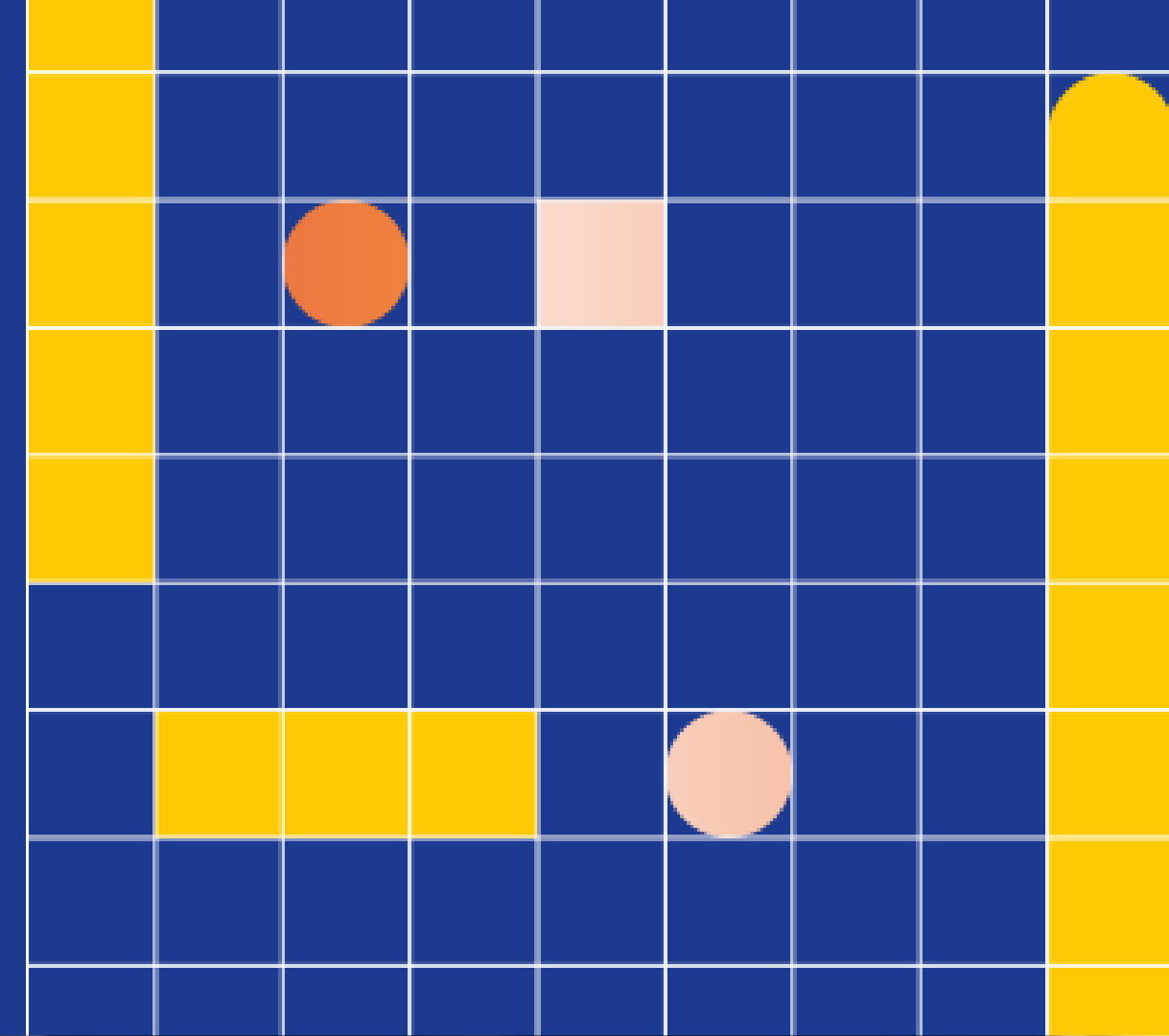
What is github?

Websites where we host repositories online



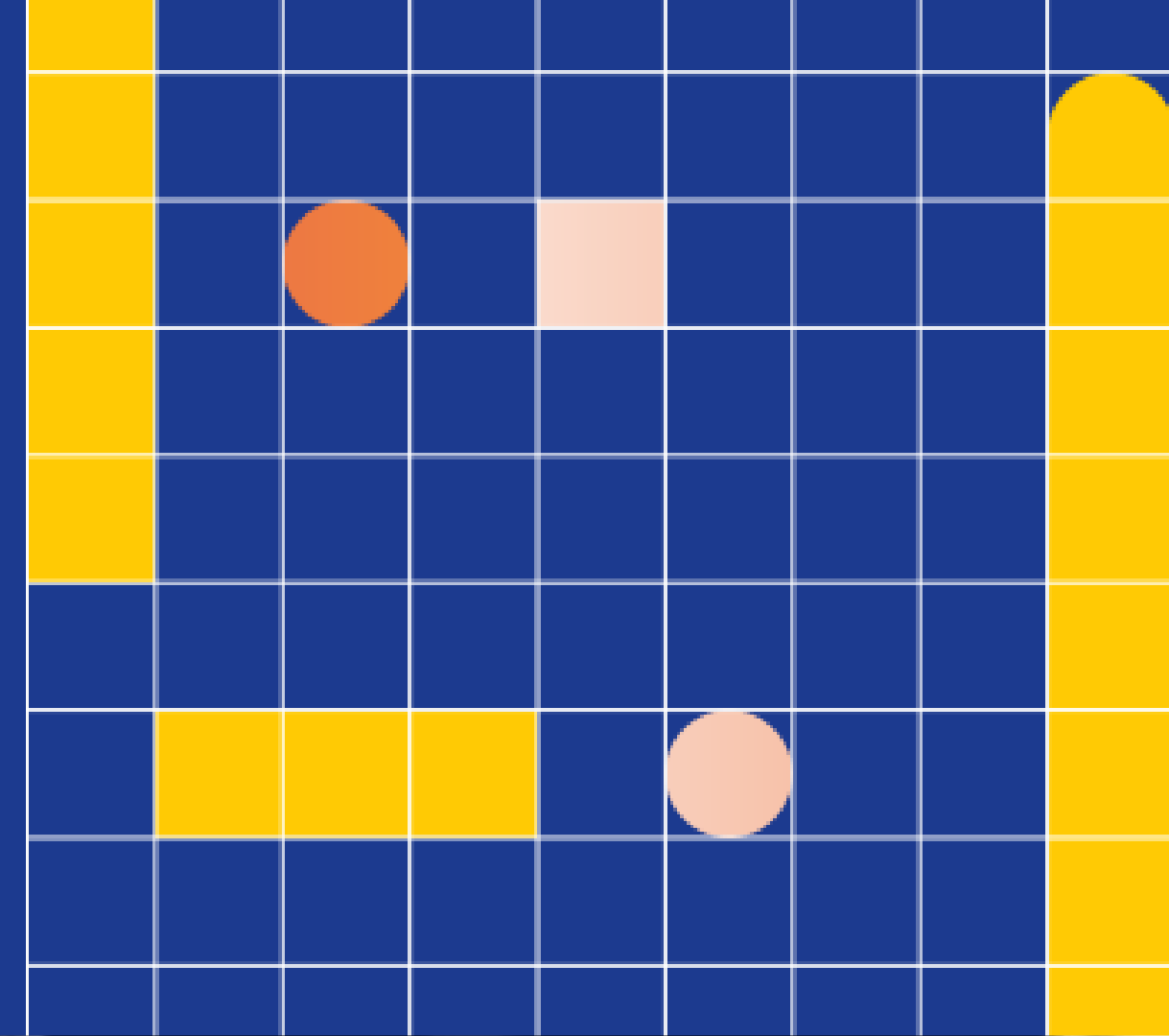
Using GitHub

- Lets create our first repository
- Write and delete content
- Check commit history



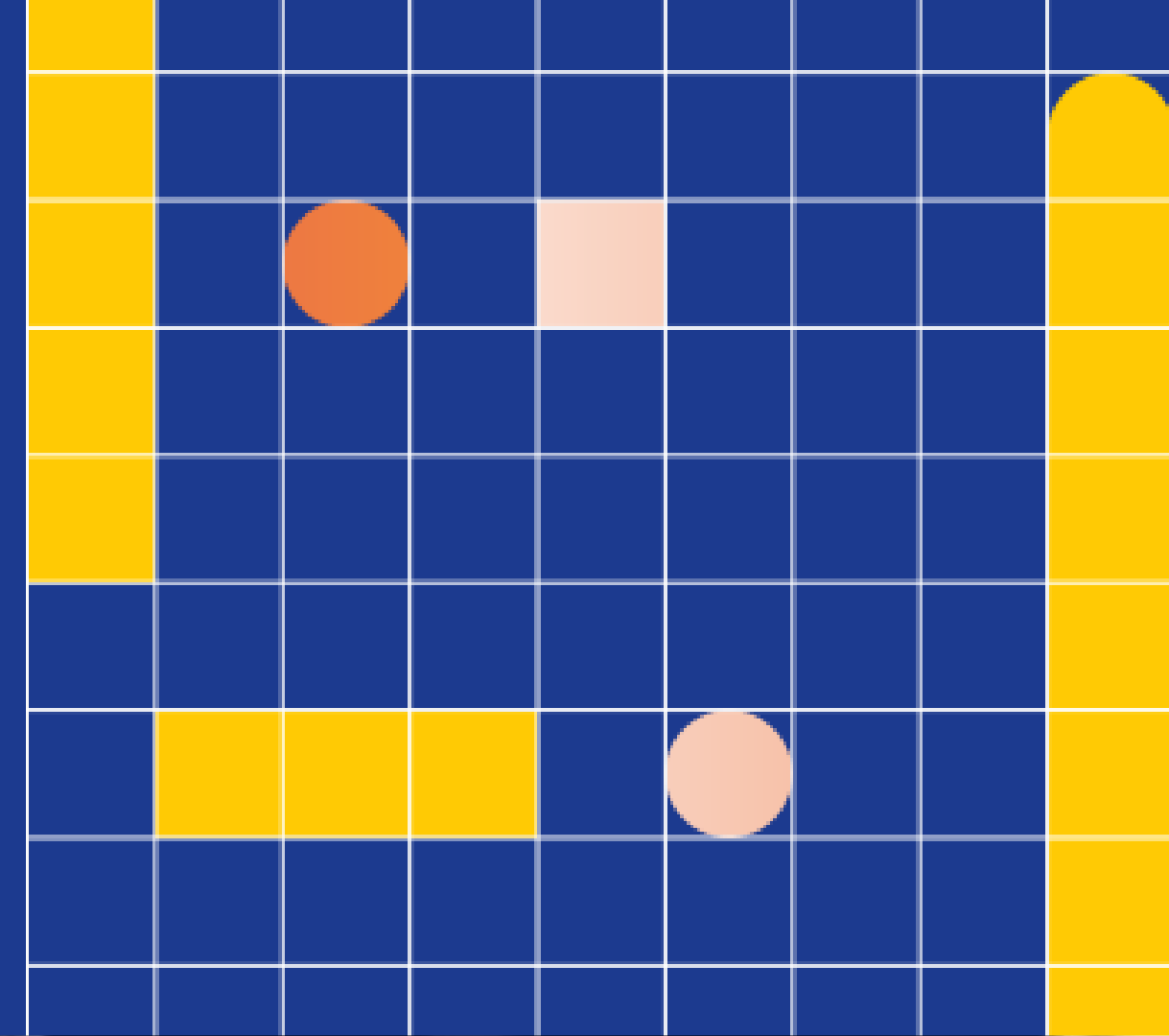
Using Git

- Command line
- IDE/Code Editors (like VS Code)
- Graphical User Interfaces (like GitKraken)



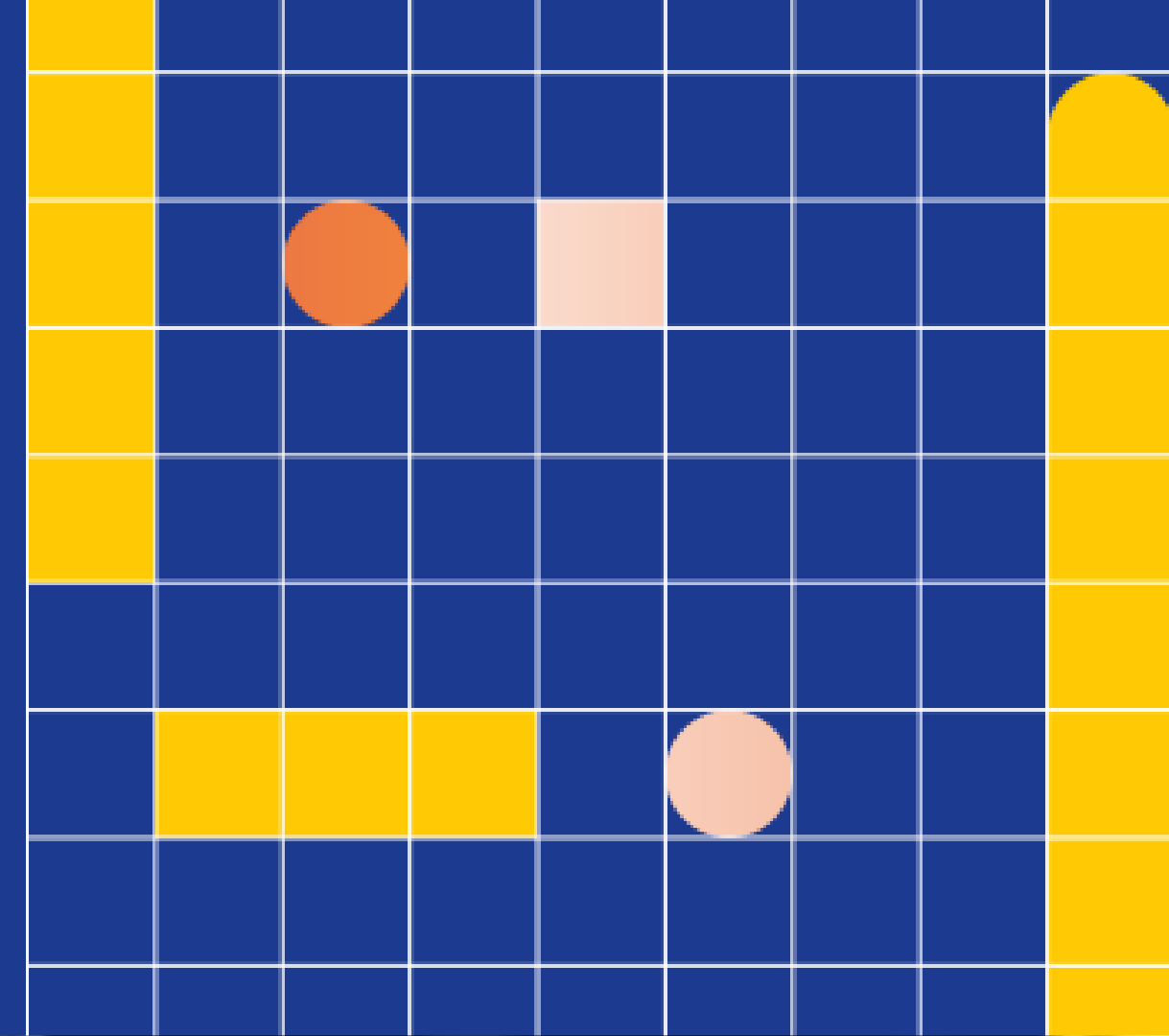
Using Git

- Git –version
- Git



Configuring Git

- Git config –global user.name “NAME”
- Git config –global user.email “EMAIL”
- Git config --list



Basic Commands

- Clone – cloning a repository on our local machine
- Git clone <-some link->

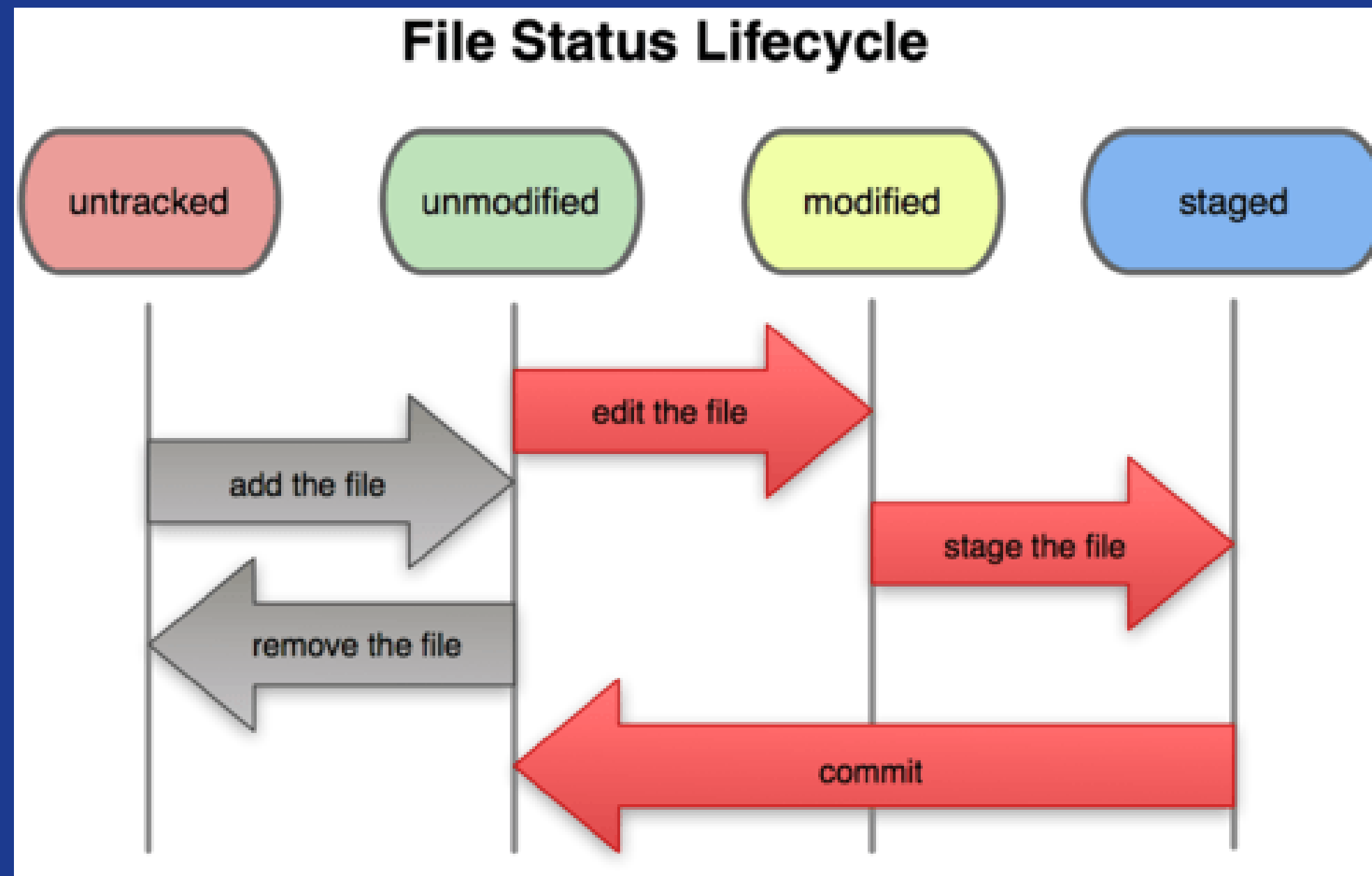


Basic Commands

- Status – displays the state of the code
- Git status



File Status LifeCycle



Basic Commands

- add – adds new or changed files in working directory to staging area
- Git add <-file name->



Basic Commands

- commit – it is the record of change
- Git commit –m “some message”



Basic Commands

- push – upload local repo content to remote repo
- Git push origin main



Init command

- Init – used to create a new git repo
- Git init
- Git remote add origin <-link->
- Git remote -v (to verify remote)
- Git branch (to check branch)
- Git branch -M main (to rename branch)
- Git push origin main

Sources



GitHub repository

Lets Connect!



Inika



Kavin Krishnan C



**THANK
YOU**